

A Short History of Artificial Intelligence

1. The Question: Can Machines Think?

The idea of building machines that think is much older than the computer. Greek myths told of bronze automatons, and in the seventeenth and eighteenth centuries inventors built mechanical figures that could write, draw or play music. These devices followed fixed mechanisms, however, and nobody seriously claimed they could reason. The modern story of artificial intelligence begins in the twentieth century, when mathematicians began to ask what it means to compute and whether reasoning itself could be reduced to the manipulation of symbols.

The British mathematician Alan Turing gave the question its most famous form. In his 1950 paper "Computing Machinery and Intelligence", Turing proposed replacing the vague question "Can machines think?" with a practical test he called the imitation game. In the game, a human judge holds text conversations with a hidden person and a hidden machine. If the judge cannot reliably tell which is which, the machine is said to have passed. Today this is known as the Turing test. Turing also predicted that by the end of the century people would speak of machines thinking without expecting to be contradicted, and he answered many objections that critics still raise today.

2. The Birth of a Field

Artificial intelligence became an academic discipline at a summer workshop held at Dartmouth College in 1956. The workshop was organised by John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon. The proposal for the workshop, written the year before, stated that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. John McCarthy coined the term "artificial intelligence" for this proposal, and the name has been used ever since.

The years after Dartmouth were full of optimism. Programs were written that proved theorems in logic, solved algebra word problems and played checkers well enough to beat their own authors. In 1958 the psychologist Frank Rosenblatt introduced the perceptron, a simple model of a neuron that could learn to classify patterns by adjusting numerical weights. Newspapers reported that such machines would soon walk, talk and be conscious of their existence. Between 1964 and 1966 Joseph Weizenbaum at MIT wrote ELIZA, a program that imitated a psychotherapist by rephrasing the user's own sentences as questions. Although ELIZA understood nothing, many users became emotionally attached to it, which surprised and worried its creator.

3. The First AI Winter

Early successes on small, carefully chosen problems did not scale to the real world. Computers were slow, memory was expensive, and many problems turned out to require an enormous amount of common-sense knowledge that nobody knew how to encode. In 1969 Minsky and Seymour Papert published the book "Perceptrons", which showed that a single-layer perceptron cannot learn even simple functions such as exclusive-or. Research on neural

networks slowed sharply for more than a decade.

In 1973 the Lighthill report, written for the British Science Research Council, judged that AI had failed to achieve its grand objectives. Funding was cut in the United Kingdom, and American agencies also became more cautious. This period of reduced funding and interest, which lasted through most of the 1970s, is now called the first AI winter.

4. Expert Systems and the Second Winter

AI returned to favour in the 1980s through expert systems. Instead of trying to create general intelligence, an expert system captured the knowledge of human specialists in a narrow domain as a large set of if-then rules. One of the most successful was XCON, used by Digital Equipment Corporation to configure orders for its VAX computers; it was reported to save the company tens of millions of dollars a year. Companies around the world built expert systems, and specialised Lisp machines were sold to run them.

The boom did not last. Expert systems were expensive to build and maintain, could not learn, and failed in strange ways when faced with situations their rules did not cover. In 1987 the market for Lisp machines collapsed as cheaper desktop computers became powerful enough to run the same software. Funding dried up again, and the late 1980s and early 1990s became known as the second AI winter.

5. The Return of Neural Networks

While symbolic AI struggled, a quieter revival of neural networks was under way. In 1986 David Rumelhart, Geoffrey Hinton and Ronald Williams published an influential paper showing how the backpropagation algorithm could train networks with several layers of neurons. Multi-layer networks could learn the exclusive-or function and much more, answering the criticism made in "Perceptrons". During the 1990s Yann LeCun and his colleagues used convolutional neural networks to read handwritten digits on bank cheques, one of the first commercial uses of the technology.

Games provided some of the most public milestones. In 1997 IBM's chess computer Deep Blue defeated the world champion Garry Kasparov in a six-game match. Deep Blue relied mainly on specialised hardware that searched around two hundred million positions per second rather than on learning, but its victory was seen around the world as a sign that machines could outperform people at a task long associated with intelligence.

6. The Deep Learning Revolution

Three developments came together in the late 2000s: far larger datasets, much faster hardware in the form of graphics processing units, and better training techniques. The ImageNet dataset, containing over fourteen million labelled images, became the standard benchmark for image recognition. In 2012 a deep convolutional network called AlexNet, created by Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton, won the ImageNet competition with an error rate far lower than any previous system. The result convinced most of the field that deep learning was the way forward.

In 2016 AlphaGo, a program developed by DeepMind, defeated the Go champion Lee Sedol by four games to one in Seoul. Go has far more possible positions than chess, and many experts had expected such a victory to be at least a decade away. AlphaGo combined deep neural networks with tree search and learned partly by playing millions of games against itself.

7. Transformers and Language Models

Language was one of the hardest problems for AI because meaning depends on long-range context. Recurrent neural networks read text one word at a time, which made them slow to train and forgetful over long passages. In 2017 researchers at Google published the paper "Attention Is All You Need", which introduced the Transformer architecture. A Transformer uses a mechanism called self-attention to let every word in a sentence look directly at every other word, and it can be trained in parallel on very large amounts of text.

In 2018 Google released BERT, a Transformer that was pre-trained on large amounts of unlabelled text and could then be fine-tuned for tasks such as classification or question answering. Fine-tuned on the Stanford Question Answering Dataset, known as SQuAD, BERT answered reading-comprehension questions about Wikipedia passages more accurately than the human baseline. The second version of the dataset, SQuAD 2.0, added more than fifty thousand unanswerable questions so that models also had to learn when the passage does not contain an answer. RoBERTa, released by Facebook AI in 2019, improved on BERT by training longer on more data.

In 2020 OpenAI described GPT-3, a language model with 175 billion parameters that could write essays, translate text and answer questions after seeing only a few examples. Two years later, in November 2022, the release of ChatGPT brought conversational language models to hundreds of millions of people and started a new wave of investment and public debate about the future of artificial intelligence.

8. Looking Ahead

The history of AI has alternated between periods of excitement and disappointment. Each wave produced lasting tools even when its biggest promises were not met: search algorithms, rule-based systems, and statistical learning are all still in use. Modern systems raise new questions about reliability, bias, energy use and the effect of automation on work. Whatever comes next, the question Turing asked in 1950 remains a useful guide: rather than arguing about definitions, we can judge machines by what they are actually able to do.